Project Prism

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Background Summary

Our team was tasked with drafting an estimate of the necessary networking equipment to allow for all events within a competition area to be live streamed to fans and competitors attending the sporting games. This is a 10-acre square area and will be hosting a state-wide gaming competition. It is in a grassy area outside of a mid-western city in the U.S with road entrances at each side. In the area, there will be different venues for the different sporting events. For track and field events, there will be a temporary oval stadium with Bleachers. The soccer, hockey, and football fields will have bleachers as well. Temporary courts with lawn areas for spectators will be available for basketball and tennis.

For the events to be carried out with as little hiccups as possible, a reliable and high-performance wireless network is required. This network must support at least 10 simultaneous live stream events at once, about 1 cellular device per person and digital ticketing to allow each person to have access to free Wi-Fi once entrance fees are paid. Cellular towers for all major carriers (Verizon, T- Mobile US, AT&T and Boost Mobile) are placed around the edges of the area to allow full cellular coverage. Spectators will be able to stream events on 1 device at a time and be able to use their cellular network to stream for a limited time per day if needed.

To fulfill the requirements, we recommend deploying an infrastructure of Wi-Fi 6 or better across the venue. This will include approximately 30 Industrial grade Access Points, supported by switches, routers and redundant copper uplinks. A captive portal/ splash page will be used to link Wi-Fi access to ticket entries. This will allow for security enhancements and bandwidth optimization tools to ensure stable connections during peak usage. Each piece of equipment will need to be carefully selected to make sure the hardware are compatible with each other and each software can support the devices selected. By doing this, the events, current and future needs will be met.

Section 1: Venue Layouts and Assumptions

Size, Placement and Distance Between Venue

On the 10-acre square land, a total of 8 fields and courts combined, and 16 6-row bleachers evenly divided between each venue and lawn areas divided among the courts. Two fan areas which include rest rooms, food area and network operation center. In total an estimate of approximately 8.82 acres will be occupied.

Below is the estimated amount of space each venue will cover:

Sub Venues	Quantity	Acres
Soccer field	1	1.86
Football field	1	1.32
Field hockey	1	1.24
Track and field	1	1.1
Tennis	2	0.34
Basketball	2	0.24
Fan area	2	2

Figure 1. Sub Venues

Each bleacher consists of 6 rows with a measurement of 33ft x 15ft and can seat about 100 people at a time. With 4 bleachers at each venue a total of 400 spectators will be able to watch each field event in person totaling 1,600 spectators. The bleachers together will occupy 0.18 acre. For court events, there will be lawn areas designated for each court seating roughly 400 spectators. The lawn area combined will cover 0.72 acre. An estimated number of 3,200 spectators should be able to watch the events in person based on the information provided above.

Distance between venues

Basketball court – Basketball court = 12 feet

Tennis court -- Tennis court = 18 feet

Tennis court 1 – Northern Fan Zone = 20 feet

Northern Fan Zone – Basketball court 2 = 20 feet

Basketball Court 2 – Hockey Field = 4,000 square feet for the Lawn area and an additional 20 feet for the walkway
Hockey Field – Track and field = 50 feet
Track and field – Southern Fan zone = 50 feet
Southern Fan Zone – Football Field = 50 feet
Football field -- Soccer field = 100 feet
Soccer field – Hockey Field = 150 feet
Soccer field – Tennis court 1 = 4,000 square feet for the lawn area and an additional 20 feet for the walkway.

The diagram below demonstrates the placement of each venue.

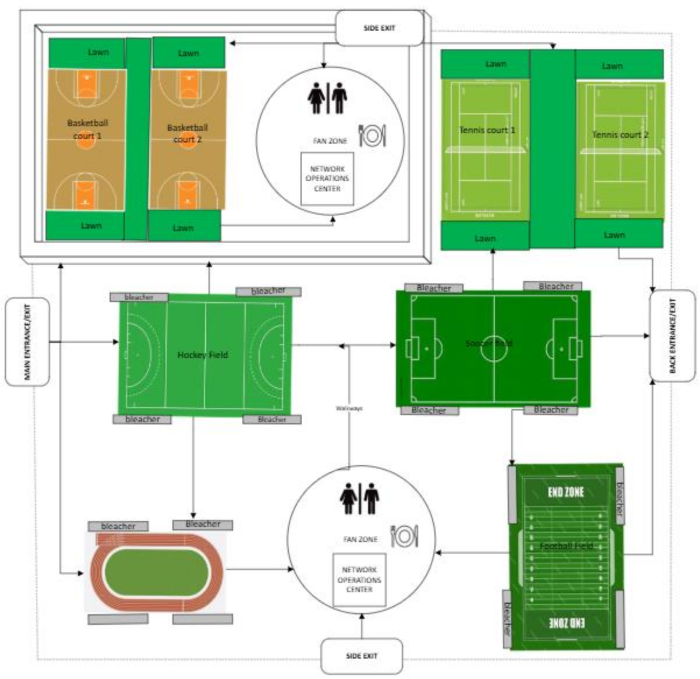

Figure 2. Place of each Sub Venue

To allow for cellular coverage throughout the area, the major cellular companies will have towers placed around the edges of the compound. With each tower needing about .23 acres, 1 tower for each major cellular company will be placed around the edges.

Figure 3. Cell Towers

The following diagram shows a detailed layout of the area with the cell towers and access points placed.

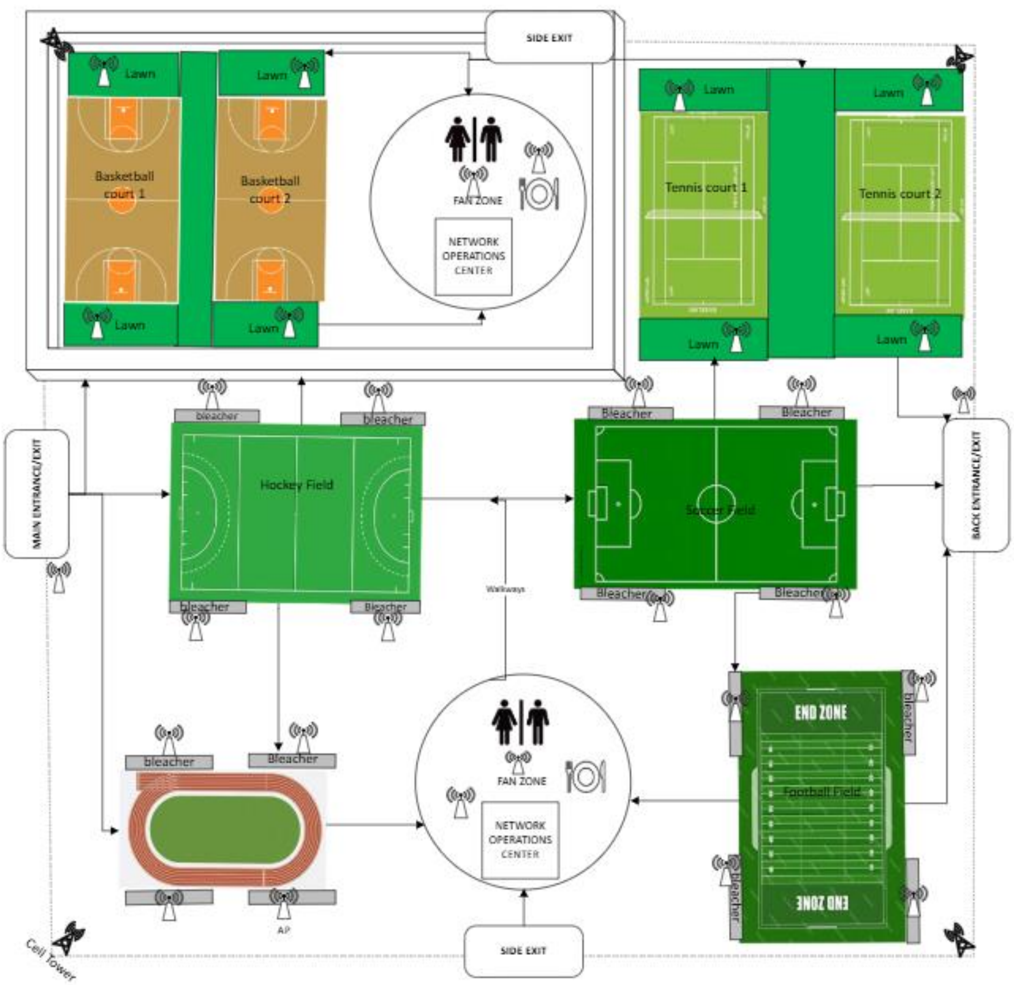

Figure 4. Detailed layout of the venue

Network Design Diagram

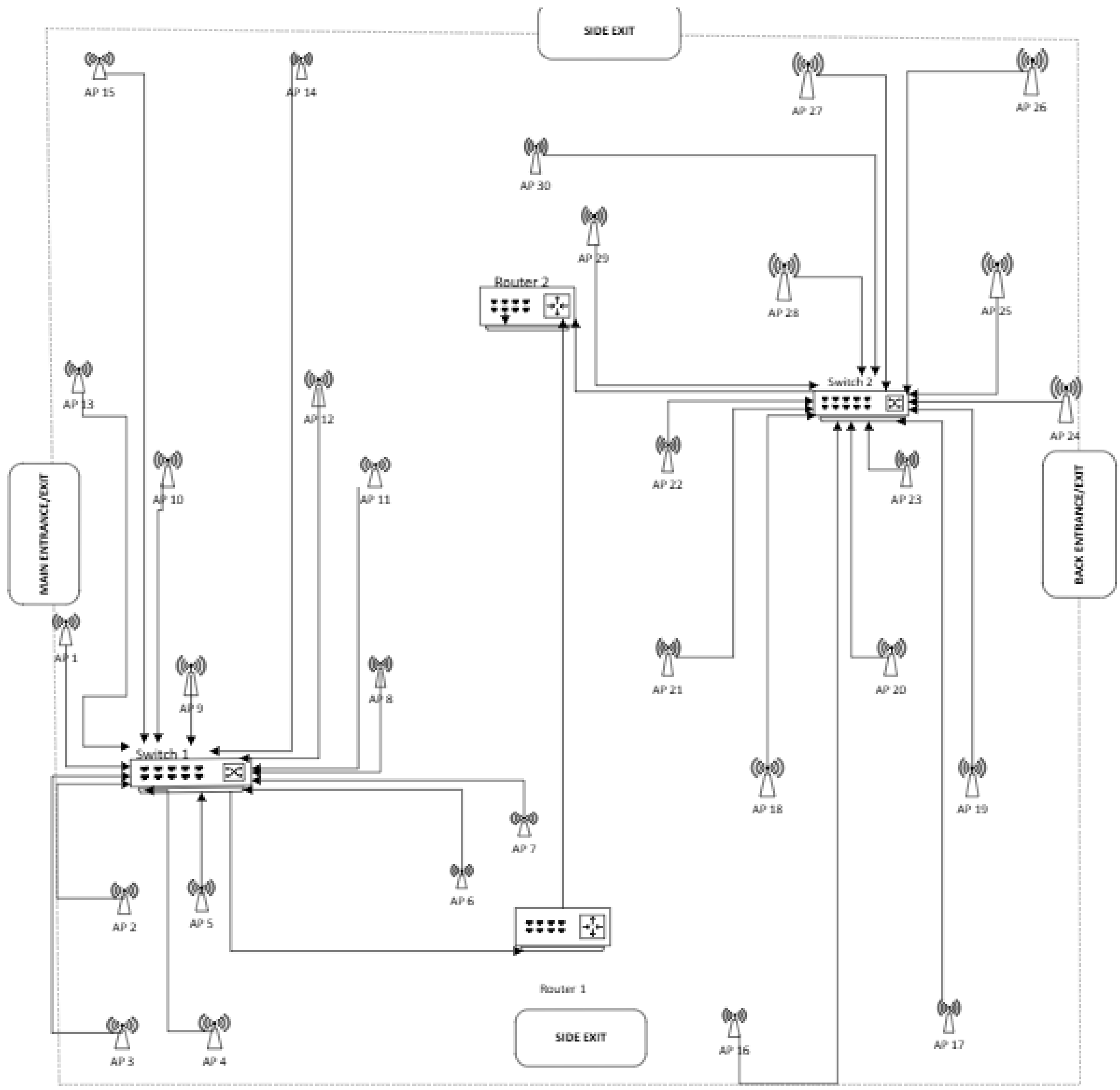

Figure 5. Network Design

The Diagram above shows the connections between access points, switches and routers. Each router is connected to one Switch, and each switch is connected to Fifteen access points (APs). Router 1 is connected to Switch 1 which is connected to APs 1-15 and services the West portion of the compound. Router 2 is connected to Switch 2 and services the East portion of the compound. Switch 2 is connected to APs 16-30. This design is based on wireless and client to server network architecture types with a Star topology. VLAN will be used to separate user groups and applications.

Section 2 – Hardware and Software

Our Team as decided to use Cisco Systems INC. devices for all networks. Using Cisco Catalyst, Switches and Access Points to carry out robust high speed wireless connection throughout the venue. Ensuring that everyone attending receives the best internet connectivity and streaming quality from anywhere on the 10 Acre land. Not only will the clients be happy, but with the affordable prices of each piece of equipment, the host will be pleased as well.

Cisco was launched in December 1984 by 2 Standford University computer scientists and went public in February 1990. Since then, the company has become a major player in the technology industry. By keeping all major hardware in the cisco family, managing and tracking becomes easier. Using the Cisco DNA Center Management Software, we can easily track and see the AP density and output of each point.

Hardware and Software Needed

Hardware/Software	Name	Quantity
Router	Cisco Catalyst 8300-1N1S-6T	2
Switch	Cisco Catalyst IE9300 Rugged Series IE-9320-24P4X	2
Access Point	Cisco Wireless 9179F Access Point	30
Server	Dell PowerEdge R750	2
Management	Cisco DNA Center	N/A
Guest Wi-Fi	Captive/Splash Page	1
Antenna	Omni Directional Antenna AIR-ANT2547VG-NS (=)	132
Surge Protection	UL 1449	1-30
Video Transmision	Teradek	1
Streaming Platform	OBS Studio	N/A
Streaming Platform	Wowza	1
Streaming Platform	vMix	1
Access control	Okta	1
Baseline Testing	Speedtest CLI	N/A
Audio management	ASIO	N/A
Diagnostics	WireShark	N/A

Diagnostics	Embedded Packet Capture	N/A
Diagnostics	Zabbix	N/A
Cable	Cat6 Direct Burial Ethernet Cable Shielded	6 (1000ft)
Cable Troughs	3 Channel Heavy Duty Cable Protector	200
Backup Power	Ventev	2

Figure 6: Hardware and Software List

Subscriptions and Licenses

To ensure smooth and professional broadcast for the sports game, our team will be using a variety of tools for network monitoring, stream management, and security.

Network Auditing and Baseline Testing

Before we go live, the team will establish a reliable network baseline using Speedtest CLI. This free tool will help us measure internet performance, including upload and download speeds, latency, and packet loss. It's an essential step to ensure our connection is strong enough to support a high-quality live stream.

Wireless Video Transmission

For video transmission, the team will use Teradek, a professional-grade wireless video solution. It allows us to send high-quality video wirelessly with minimal delay. The team will also be using Teradek Core, their cloud-based management service, which helps us control and monitor streams remotely. Teradek is not free. The Core 3.0 Pro business package starts at approximately $3,500, with pricing depending on the features we need.

Streaming Platform Setup

After establishing our network baseline, the team will stream through Wowza, a paid streaming platform that supports multi-destination broadcasting, live webinars, and highresolution video playback. This will give us flexibility and reliability when reaching our audience who will be watching our live stream.

To handle video production, the team will pair Wowza with OBS Studio (a free, widely used broadcasting tool) and vMix (a paid software that offers more advanced live production features). Together, these tools will allow the team to manage multiple video sources, apply graphics, and deliver a polished broadcast.

Security and Access Control

To secure our live stream and prevent unauthorized access, the team will integrate Okta. This is an AI-powered authentication platform that helps the team manage who can access the stream and adds protection against unwanted interruptions. Pricing starts at around $6 per user/month, with additional costs depending on the level of features the team chooses.

Audio Management

The team will use ASIO, a low-latency audio driver, to help manage sound quality during our broadcast. This ensures audio stays in sync with video and meets production standards. Costs vary depending on whether the team chooses to use standard or professional-grade audio equipment.

Monitoring and Diagnostics

To monitor system performance and quickly identify any issues, the team will utilize Zabbix and Wireshark, both free tools. Zabbix will help the team track system health and performance, while Wireshark will assist us for deeper network diagnostics when needed.

Network Infrastructure Management

For network traffic management and stream stability, the team will be using Cisco DNA Management. This software is a reliable enterprise networking solution that helps us manage bandwidth, prioritize streaming traffic, and maintain consistent performance. The platform requires per-device subscription licenses, which the team will need to factor into for planning and budgeting.

Backup Power

Our choice of backup for this event is the Ventev outdoor UPS system, and it is a critical component in ensuring an uninterrupted power supply during the sports event. This system is designed specifically for outdoor use, housed in a NEMA 4X polycarbonate enclosure, making it highly durable and resistant to any environmental factors that may occur during the event. Its design suits outdoor network IoT applications, such as security, monitoring, and edge computing storage, providing a reliable power source for critical devices even when the main power supply fails.

This UPS system converts 120 VAC input power to 12 VDC output, suitable for powering active equipment commonly used in event setups, including network devices, Wi-Fi radios, and low power PoE devices. This system supports efficient power conversion and delivery, making it adaptable to the variety of connectivity and communication technology we placed throughout the venue. An important feature of the Ventev outdoor UPS system is its field configurability, which allows easy integration and customization to meet specific deployment requirements. This flexibility is important in a dynamic environment where power demands may vary, and rapid adjustments may be needed. This system can be mounted on poles or walls, which can optimize space and installation efficiency around the sports venue.

Section 3 Wi-Fi Planning

The driving force behind the Wi-Fi network will be a pair of Cisco Catalyst 8300-1N1S-6T enterprise level routers. This specific model was chosen over similar Cisco routers due to its

6 x 1G WAN instead of a singular 10G WAN. The 6 different bands will be much more suited to handle the load of lots of spectators, which is exactly what is needed in a large sports venue like ours. Included is an integrated firewall and many other features that will improve the experience of users. In case of a power outage or disruption, the routers have two power supplies, allowing one to be connected to a backup. Power backups will be mentioned below.

Figure 7: Cisco Catalyst 8300-1N1S-6T

As the Cisco Catalyst 8300-1N1S-6T router only has so many ethernet ports, the number of Wi-Fi access points is limited. However, by connecting some network switches, we will be able to bypass this limitation. Network switches are useful as they can expand a singular ethernet port into many more, which is perfect for lots of access points. The Catalyst IE9300 Rugged Series IE-9320-24P4X is an industrial switch made for deployments that require high-density PoE, easy scalability, and traffic aggregation. It has 24 GE copper ports with PoE+, 4 10-GE SFP+ ports and 720W of total PoE power budget and is ideal for outdoor and inclement conditions. One port is used for the router source, and fifteen more are used to then connect to the Wi-Fi access points. This switch runs Cisco IOS XE which is an operating system with built in security and trust, it also includes Embedded Packet Capture (EPC) which is needed for the device to communicate with WireShark.

Figure 8: Catalyst IE9300 Rugged Series IE-9320-24P4X

The final aspect of the basic network setup is the Wi-Fi access points (APs). Wi-Fi access points serve to connect the guests to the service over a large area. Twenty of these access points will be scattered over the 10-acre venue, allowing access from any location, and support for the many clients expected. Using the Cisco Wireless 9179F (CW9179F) Access Point, a high-capacity outdoor Wi-Fi 6E, we can provide coverage to the entire area with just 30 APs. Each access point can cover up to 300ft in diameter and can comfortably host roughly 150 guests leaving virtually no dead zones in the venue. The CW9179F Firmware includes sniffer mode allowing the APs to stream packets to a PC with WireShark or WLC. Having 5-10% extra APs for backup is also recommended. For more coverage, the CW9179F APs can be used with 4 external antenna each. If needed, AIR-ANT2547VG-NS (=) Antennas can be paired with this device and prices can be checked on an as needed basics since the number of antennas that would be needed equals 120 and an additional 12 on hand as spares.

Figure 9: Cisco CBW150AX Wi-Fi 6 2x2 1 GbE Access Point

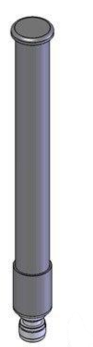

Figure 10: Antenna: AIR-ANT2547VG-NS (=)

With routers, switches, access points, and cabling determined, we can get an idea of where everything will be placed. Routers are placed within the data centers, and switches are placed strategically to each handle of 15 access points. Nothing can be placed on an actual field, as it would get in the way of games being played. Placing access points in the middle of fields would also be redundant as no fans watching the live streams are on the actual field. Also, with 10 acres being a total of 435,600 square feet, each access point would need to cover an area of 17,000 square feet for everything to be covered with no redundancy. The access point's range of 300 feet diameter gives a coverage area of 70,000 square feet, leading to plenty of overlap to cover the entire venue and reduce a concentrated load if needed.

Before the routers can broadcast the stream, servers are needed to manage all the hardware and incoming video. With the scale of streaming desired, an enterprise level server will be required. A Dell PowerEdge R750 server rack will house the hard drives and processors. Since a server level processor is needed, the Intel Xeon line is perfect, specifically the Intel® Xeon® Gold 6338 2G, 32C/64T, 11.2GT/s, 48M Cache, Turbo, HT (205W) DDR4-3200. This processor is provided as an option straight from Dell and thus will work together.

Finally, solid state drives and hard drives are both needed. A smaller 2 TB solid state drive will be the fastest of the two and thus will serve to drive the operating system as well as live streams. A total of four 2.5 TB hard drives will also be used to store old streams, totaling 10 TB of storage for this purpose.

Figure 11: A Dell PowerEdge R750 server

Figure 12: Intel® Xeon® Gold 6338 2G, 32C/64T, 11.2GT/s, 48M Cache, Turbo, HT (205W) DDR4-3200

Guest Network Setup

The venue will operate a main network for employees and network administrators as well as a guest network for guests and athletes. For spectators, once they purchase a ticket they will be asked to provide an email address to receive the digital ticket. That same email used when purchasing the ticket will be the user ID for Clients when they are ready to use the Wi-Fi. For athletes, once they have registered to participate in the event they will be emailed a user ID that will allow them to access the internet. This network will serve as free Wi-Fi and will also have access to live streams and past events while connected.

Clients will be able to connect to the internet using a Captive portal/Splash page before being able to access the guess Wi-Fi. This page will use a HTTPS browser to ensure maximum security. Guests using this connection will need to interact with the page. Not only that, but a splash page is a great way to advertise events and allow guests to go directly to the streaming page without a hassle. Each guest is allowed to connect one device under their user ID. To stream on multiple devices, they will be able to connect to their cellular network Wi-Fi for a limited amount of time.

Cabling and Infrastructure Estimates

The cabling for the venue will require a substantial amount of outdoor rated Category 6 copper ethernet cable. With 2 switches and 30 Wi-Fi access points, a total of 32 cable sections will be necessary. Outdoor rated Cat6 ethernet cables have a maximum length of 100 meters or

328 feet, and this maximum will be needed for some sections. To obtain a conservative estimate, an average cable section length of 200 feet is assumed.

Figure 13: Cat6 Direct Burial Ethernet Cable Shielded

When routing the cables through the venue, some more items are needed to protect them from the elements as well as foot traffic. As found earlier, 4800 feet of total ethernet cabling is required. We need approximately 200 3 channel heavy duty cable protectors to cover all the cables above ground. To save money, half of the cabling will be covered, specifically in the highest traffic areas. This amounts to 2400 feet. Heavy duty cable troughs are readily available solutions to protect cables, and while expensive, are a worthwhile investment.

Figure 14: 3 Channel Heavy Duty Cable Protecter

Security

Ensuring security of the Wi-Fi network at the sports event is a crucial component for the event. The infrastructure is based on IEEE 802.11ax, or Wi-Fi 6 technology, designed to deliver faster speeds, greater capacity, improved efficiency and enhanced security. The deployment includes approximately 30 industrial-grade Cisco Wireless 9179F Access Points (APs), which we strategically placed across the venue to provide more robust coverage. These APs incorporate security features such as WPA3 encryption and enterprise-grade authentication protocols, such as 802.1X, helping to prevent unauthorized access and secure wireless transmissions.

The Cisco Catalyst 8300-1N1S-6T router is central to the network. It manages traffic flow and enforces firewall policies to isolate guest traffic from sensitive operational data. The network's resilience is enhanced by the Catalyst IE9300 Rugged Series IE-9320-24P4X switches, which are designed for outdoor conditions with high PoE+ power budgets, support for rapid traffic aggregation and segmentation. The design of these switches ensures continuous secure operation despite possible environmental challenges such as rain, lightning, and UV exposure.

In addition to hardware, the Wi-Fi network leverages software tools to strengthen security and streamline management. The Dell PowerEdge R750 server plays a vital role in hosting network authentication services, logging into security events, and supporting captive portal functionality that links Wi-Fi access to ticketing systems. This captive enforces user authentication and acceptance of terms of service, which effectively controls access and reduces the risk of unauthorized connections.

Network management is centralized through Cisco DNA center, which provides us with real-time monitoring, automated policy enforcement, and rapid threat response capabilities. This enables the IT team to quickly isolate threats, manage bandwidth allocation, and maintain quality of service during peak usage times. Together, the combination of our selected hardware and software creates a secure, high-performance Wi-Fi environment suited for the high population density and dynamic nature of this sports event.

Section 4 - Implementation Plan

Timeline

Step one: Strategy and Proposal

Step two: Fine Service Level Agreements

Step three: Purchase and Test devices for quality Assurance

Step four: Deploy design and Perform field testing

Step five: Make adjustments and perform final testing

Step six: Handover to onsite management

Estimated Cost

Price of Each Component

Name	Quantity	Cost
Cat6 Direct Burial Ethernet Cable Shielded	6	$1,596.00
Cable Troughs	200	$ 84,000.00
A Dell PowerEdge R750 server rack, Intel®Xeon® Gold 6338 2G, 32C/64T, 11.2GT/s, 48M Cache, Turbo, HT (205W) DDR4-3200	1	$11,600.00
Cisco Catalyst 8300-1N1S-6T	2	$12,000.00
Catalyst IE9300 Rugged Series IE-9320-24P4X	2	$17,236.00
Omni Directional Antenna AIR-ANT2547VG-NS (=) (Optional)	132	Varies
Cisco Wireless 9179F Access Points	30	$151,980.00
Cisco DNA Center	Subscription	$ 4,500.00
Teradek	Subscription	$3,500.00
OBS Studio	Subscription	Free
Wowza (Pay as you go plan)	Subscription	$25-1,560.00
vMix	Subscription	$50-1,200.00
Okta	Subscription	$6.00
Speedtest CLI	Subscription	Free
ASIO	Subscription	Varies

WireShark	Subscription	Free
Zabbix	Subscription	Free
UL 1449	1-30	$100+
Ventev Outdoor UPS System	2	$3,902.00
Total		**$218,678.00**
Maintenance and reserve budget		**At least $50,000.00**

Figure 15: itemized list and cost

Section 5 – Potential Problems and Solutions

Problems and solutions

The following set of issues has been identified and prepared to troubleshoot potential disruptions at every layer of the network that could interfere with the live streaming of the sports competition. These scenarios, ranging from minor to critical, have been carefully categorized to ensure that every possible situation is anticipated and addressed in advance. The goal is to implement preventative measures and provide effective solutions before problems arise.

Below is an organized list of potential issues and their corresponding troubleshooting strategies for the existing network infrastructure.

Environmental issues

During the sports game that is held with outdoor venues, various physical layer vulnerabilities can arise that may even compromise the performance of the network. When weather conditions allow the sports activities to proceed, they may still introduce challenges that hinder the effective streaming of a live game.

Network cabling is typically the most vulnerable component in such a setup. It can be compromised by both environmental factors and physical interference. Water or moisture intrusion, resulting from rain, humidity, or condensation, can lead to signal degradation, corrosion, or even short circuits. These issues can manifest as reduced bandwidth or intermittent connectivity of the live stream during the sports game.

In addition, power outages caused by storms can result in connectivity loss or even hardware damage. Even brief interruptions may significantly degrade overall network performance and disrupt the live stream of the sports game.

Extended UV exposure, followed by sudden shifts to rain or lightning, can further deteriorate cabling. Prolonged exposure can break down cable sheathing, while lightning events pose direct risks of power surges, equipment failure, or fire hazards. Material fatigue due to fluctuating temperatures and weather patterns may also weaken the cabling infrastructure over time, reducing its effectiveness and lifespan.

Furthermore, natural surroundings such as local wildlife, vegetation, building integrity, wall materials or root growth can pose threats to buried or exposed cables. Repeated environmental interactions combined with foot traffic from event staff or attendees increase the likelihood of cables being damaged, cut, or disconnected, resulting in compromised physical connectivity.

Environmental Solution

These issues could collectively happen at once or present as an issue throughout the day of the sports game event. In the probability that any of these cases may occur, the immediate action taken by the team will be the following:

In preparation for weather conditions that could compromise cabling and affect the efficiency of streaming during the event, the team will implement several protective measures to ensure the integrity of equipment and minimize service disruption.

To safeguard all physical equipment from hazardous electrical issues caused by lightning or static discharge, surge protectors (UL 1449) will be strategically installed. These will mitigate the risk of electrical surges during the sports game or prior due to atmospheric conditions. This protection will cover all critical components including switches, PoE injectors, and outdoor radios.

All equipment used in outdoor conditions will be housed in NEMA 4X-rated enclosures and connected using IP68-rated cabling to guard against water ingress, dust, and corrosion. The use of gel-filled Cat6 cabling with UV resistance and temperature tolerance ensures durability in various environmental conditions. This with our team's staff doing rounds to monitor all hardware during non-optimal conditions will also be in effect.

Additionally, Ventev outdoor UPS systems will be used to provide backup power to PoE-enabled devices. Installation will occur before the event unless such systems are already in place, in which case functionality will be verified during inventory and asset management. UPS efficiency will be calculated using the power/wattage of devices used added with a buffer to help account for any overloading issues in the case that the original power is compromised.

Combined with this the location of cables for outdoor runs and a map (see page 5-7) of those runs will include markers of each cable for that may run outside in case of weather events that could cause a power outage or equipment issues and to cover the avoiding of cuts, digging or physical injury to the cables. Routine inspections from the team of these areas will be done to further prevent loose connections or corrosion.

Critical to our network performance is the power source to each of our physical hardware devices. This means that, with the calculation of the total wattage used for this event and an added buffer for surge loads by the team, a generator backup will be in place using an automatic transfer switch (ATS) to monitor the voltage and frequency in the event the main power is lost. The equipment determined for this would be an Eaton ATS16Netpack. When this is detected, the ATS will switch to the generator without the need for any manual switching. Ideally, the ATS will be closed transition in case of emergencies.

Considering the building structure and possible degradation of the live stream of the sports game, the degree of attenuation must be accounted for. A survey of the area would be conducted prior to event to collect access points and effects of the material between each point. To resolve the increased latency issues and potential dropouts, access points will have to be placed to bypass these obstructed areas. The use of mesh nodes for concrete possibility and lower frequency bands (2.4ghz for lower throughput) will be implemented. Access points will be arranged by competing devices, distance, material and shared bandwidth based on these additional considerations to help prevent streaming issues.

All outdoor runs will terminate in a centralized, environmentally controlled equipment room, which will house the core network infrastructure. Buried cables (Cat6 or shielded Cat5e and Ubiquiti ToughCable Pro) will be used to improve resilience against weather and physical damage. In collaboration with this, monitoring to gather a baseline assessment is necessary to apply to conditions and react in accordance with troubleshooting. This should also rule out any overlapping AP or if insufficient bandwidth occurrences due to high density use on a single AP.

Software and upload issues

With provisions made for the physical layer of the network, it is expected that software issues could impede any progress made. The reliability of the hardware is dependent on the possibility of catching these scenarios as they occur. With this, the combination of both software and hardware come the possibility of software issues.

In the effort to maintain or prevent the possibility of losing connection via hardware, it is necessary for the team to have real-time alerts monitoring the UPS, generator and the networks health. These tools provide real-time feedback in case any of the issues occur and are not visible to an outdoor assessment. This will require network monitoring tools.

In the case that all hardware remains physically fine, the possibility of software crashes is a risk during broadcasting. Failures to stream live, lag or fragmentation would result in dissatisfaction. This could also result in or be part of firmware conflicts and malfunctions. Choosing the correct platform for streaming and the right software that can ensure that viewers are able to watch a live broadcast of the sports game, which is integral to the team’s goal.

Software and upload solution

The high attendance of the sports game event may include various problems and impose real-time solutions for the proper viewing of the stream. This relies heavily on the software used to monitor, respond and troubleshoot variables not accounted for or obstructing live streaming. The team will use in correlation with our streaming software, will use Zabbix and Wireshark along with Speedtest CLI to monitor, detect and alert. Speedtest would be used for real-time network upload/download speeds and Wireshark would be implemented for analyzing packet loss or latency.

In the event that during the live streaming, the software for live streaming during the sport event is overwhelmed, there will be a backup encoder or scene switcher ready (OBS

+vMix) along with hardware encoder (Teradek) for redundancy. This should prevent the software running into any conflicts or overloading the CPU and GPU.

If there are audio or video synchronization issues or lags in the video or audio, due to software misconfiguration, the team will be utilizing OBS and vMix which will be implemented to reduce lag as well as keep audio and video synced. With the use of ASIO, the team can both secure high synchronization, throughput and low latency can be optimized. This is paired with NVENC and close monitoring of systems, so video quality is retained, and the live stream of the sport event avoids insufficient bandwidth.

Using Woza streaming cloud with OBS and using auto-reconnect settings, a delay in the stream can be avoided and will be optimal on any device or platform. This should also help avoid connection issues if the live stream cuts out for any of these reasons.

Prior to the event, the team will ensure that all systems are correctly updated, drivers and devices ensuring optimal delivery of the stream. Creating a list of all licenses and subscriptions used prior to the event and making sure all software is confirmed and activated via license should ensure that now software locks up during streaming.

Security issues

High attendance and staffing of the sports game will increase the probability of security issues which could compromise the network. This could come in both physical and network security. Tampering of devices with the control room with unauthorized access or within the network is a probable event that should be considered and prepared for.

In the event of physical tampering, this could be seen if the equipment room is not properly secured by a lock or monitored by staff. This could allow anyone during the sports game event to access the room and expose any devices or cabling damage or corruption. Without proper team identification, locks and system controls exposure could impede live streaming.

The possibility of disruption of live streaming could also happen within the system. This could look like various attack attempts, like unauthorized access from outside without access control. It could mean that traffic overload occurs on the systems from a Denial-of-Service (DoS) attack or unauthorized content injection (This could be problematic for young viewers within the audience as well). With the access to this stream, there is also the possibility of a data leakage from the lack of encrypted data which could be from Man-in-the-middle attacks.

Security Solution

The result of physical security is implemented immediately within the team. Security policies and procedures are placed to prevent disruptions along with security awareness training. This should create a list of authorized users who may move in and out of the room along with those who have access controls. To migrate risk, threats or vulnerabilities on the grounds, a network security control policy creation will create safeguards for building access, and safeguards that pertain to any of the listed equipment and programs. This will extend to the venue's building security. Having staff doing occasional rounds of exposed equipment outside the wiring closet would be included in this provision. Auditing any vulnerabilities on the venue prior to the event will also be implemented to reduce vulnerabilities.

In relation to security measures within the software, several measures should be applied. Unauthorized network access outside the area is a potential threat to the streaming of the game. To prevent this, using WPA3 encryption on all wireless access and employing MAC address lists along with having two separate Wi-Fi access for Staff and audience would be integral. The addition of SNMP, and NetFlow as software to provide real-time feedback for any anomalies in the system will help reduce the likelihood of this occurrence.

Specific attacks such as DoS and Man-in-the-middle attacks can be mitigated by ensuring data is encrypted using TLS/SSl with a VPN along with disabling unencrypted services. Configuring fire walls and having real-time monitoring using Zabbix or PRTG can be used to mitigate attacks like this that could harm the quality of service (QoS). Securing stream keys for public use and requiring token-based access in place will secure these events. Having two Internet Service Providers as a failure over due to saturation and the use of Ubiquiti for automatic WAN failover routers will add additional security to this problem.

To prevent the possibility of traffic overload during streaming, traffic shaping tools such as NetFlow and Cisco routers will help control bandwidth, prioritize traffic and reduce latency for streaming. Utilizing FortiGate firewalls and cFosSpeed will help in optimizing the streaming for an optimal user experience.

Overall, having a Computer Security Incident Response Team (CSIRT) in accordance with an Incident response plan (IRP) will ensure that every security incident is legally protected and properly controlled or investigated.

Conclusion

In conclusion, the team for these sports events thoroughly assessed the premises for all accounted gear and hardware required for the live stream of the sports event. Physical and network Security will be optimized with continuous monitoring to address potential issues. The team has optimized necessary hardware and software scaled appropriately for the size of sports games, venues, and overall layout.

With accrued potential costs accounted for, along with mapped out Wi-Fi design, access point placement, and other critical infrastructure mappings, the team is properly prepared to provide and support the live stream of the game event. With all necessary probabilities and issues addressed with planned solutions in place of rapid response, the team is confident in its ability to successfully execute the live broadcast of the event.

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